

MID SEM 2026
Foundations of Biomedical Informatics(BIO222)

Max. Marks: 100

SECTION-A (15*4=60)

1. In a screening system for early cancer detection, missing a true case is considered far more dangerous than triggering unnecessary follow-ups. Which metric should be prioritized?
A. Precision **B. Recall** C. Accuracy D. Specificity
2. In ontologies, a concept is sufficiently defined when its relationships uniquely distinguish it from its supertype and siblings. This property enables:
A. Automated reasoning B. Faster queries C. Data encryption D. Billing control
3. A researcher uses HL7 messaging to exchange patient vitals across systems, but does not standardize units. The failure lies at which level?
A. Technical B. Network C. Hardware **D. Semantic**
4. A hospital develops a mortality prediction model using ICU data. The model performs very well internally but fails when deployed in another hospital using the same ICD-10 codes. What is the most likely conceptual reason?
A. The model was not deep enough
B. Coding systems guarantee identical clinical meaning
C. Clinical context and intent behind coding differ across institutions
D. The dataset size was too small
5. A researcher randomly splits rows of MIMIC-IV data, allowing multiple ICU stays from the same patient to appear in both training and testing sets. The inflated performance primarily reflects:
A. Patient-level data leakage B. Balanced class distribution
C. Model generalization D. Statistical robustness
6. In SNOMED CT, representing “acute bacterial pneumonia of left lung” using multiple attribute-value relationships instead of a single predefined concept is an example of:
A. Pre-coordinated terminology **B. Post-coordinated expression**
C. Flat vocabulary mapping D. Billing-based classification
7. In a hospital using an electronic medical record (EMR) and a separate billing system, what is the main role of HL7 FHIR in achieving interoperability between these Hospital Information Systems (HIS)?
A. To act as a standard way to structure and exchange healthcare data via APIs
B. To provide only a visual dashboard for doctors without affecting data exchange
C. To replace all existing hospital software with a single unified system
D. To store all hospital data in a single central database managed by the government
8. In an HL7 message, which symbol is used as the component separator?
A. & **B. |** **C. ^** D. ~

Assertion-Reasoning

- A. Both Assertion and Reason are true, and Reason correctly explains Assertion
- B. Both true, but Reason does NOT explain Assertion
- C. Assertion true, Reason false
- D. Assertion false, Reason true

9. **Assertion (A):** High accuracy in imbalanced healthcare datasets may not reflect good clinical performance. **Reason (R):** Accuracy equally penalizes false positives and false negatives in all clinical contexts.
A
10. **Assertion (A):** HL7 messaging ensures shared clinical meaning across systems.
Reason (R): HL7 primarily standardizes message structure and transport format.
D
11. **Assertion (A):** A model with high precision but low recall is unsafe for disease screening tasks.
Reason (R): Disease screening tasks are primarily concerned with minimizing false positives rather than false negatives.
C
12. Errom's Law predicts that the number of new drug approvals per billion dollars spent is increasing. T/F **Answer: False**
13. Percentiles divide the data into 10 equal parts. T/ F **Answer: False**
14. HIPAA stands for Health Insurance Portability and _____ Act. **Answer: Accountability**
15. The square root of variance is called _____ **Answer: Standard Deviation**

SECTION-B (4*10=40)

1. A clinician records: "Chest pain radiating to left arm with sweating." **a)** Explain how post-coordination in SNOMED CT allows more computable meaning than pre-coordinated ICD coding. **b)** Using defining attributes and supertype relationships, explain how automated reasoning can infer "possible myocardial infarction." **c)** Why would primitive concepts weaken clinical decision support? **[3+5+2]**

Answer:

- a. ICD is a pre-coordinated classification system, meaning a single predefined code represents a diagnosis (e.g., myocardial infarction). It does not formally encode relationships between individual symptoms, such as radiation pattern or associated autonomic signs. (1)
SNOMED CT supports post-coordination, where multiple concept identifiers are combined using structured attributes (e.g., finding site, associated symptom).
Thus, Chest pain, Radiation to the left arm and Sweating can be represented as a structured logical expression. (1)
Because SNOMED CT encodes meaning using formal relationships, it provides richer computable meaning than ICD's fixed codes. (1)

Criteria	Marks
Explains ICD as pre-coordinated	1
Explains SNOMED post-coordination	1
Explains richer computable meaning	1

- b. SNOMED CT is structured using:
Supertype (is-a) relationships
Defining attributes
Description logic

Myocardial infarction (MI):

- Is-a → Ischemic heart disease
- Is-a → Cardiovascular disorder

Defining characteristics may include:

- Finding site → Myocardium
- Associated morphology → Infarction
- Associated symptoms → Chest pain
- Autonomic features → Sweating

When the clinician records:

"Chest pain + radiation to left arm + sweating"

These findings match defining characteristics of myocardial infarction. Because the concept is sufficiently defined, an automated reasoner can:

- Match symptom pattern
- Traverse hierarchy
- Infer “possible myocardial infarction”

Criteria	Marks
Explains supertype hierarchy	1
Explains defining attributes	1
Links symptoms to MI characteristics	1
Explains description logic reasoning	1
Connects to decision support	1

- c. A **primitive concept** lacks sufficient defining attributes to uniquely distinguish it from sibling concepts. Without complete logical definitions:
- Automated reasoning cannot reliably infer relationships.
 - Classification may be incomplete.
 - Clinical alerts may fail or be inaccurate.

Thus, primitive concepts weaken computable precision and reduce effectiveness of decision support.

Criteria	Marks
Correct definition of primitive concept (lacks sufficient defining attributes)	1
Clear explanation of impact on reasoning and decision support	1

2. A 55-year-old patient visits a hospital after experiencing chest discomfort. He is first given a wearable device that records continuous heart activity. The doctor then reviews his hospital records, including previous diagnoses and laboratory reports. To investigate further, a cardiac MRI scan is performed. Genetic testing is also conducted to identify inherited risk factors. Finally, the confirmed case is reported to a national disease registry for population monitoring. For each stage, write: Type of biomedical data source and Relevant data standard/format commonly used. For eg, Step 1 → Data Source: _____ | Standard/Format: _____
[5*2=10]

ANSWER:

Step 1 → Biomedical Signal Data | IEEE 11073 / Sensor data formats

Step 2 → Clinical Data | HL7 / FHIR / SNOMED CT / LOINC

Step 3 → Imaging Data | DICOM

Step 4 → Genomic Data | FASTA / FASTQ / GenBank format

Step 5 → Public Health Data | ICD / HL7 reporting standards

3. A large tertiary hospital is facing increasing antibiotic resistance cases, leading to longer hospital stays and higher costs. The hospital wants to transform itself into a Learning Health System (LHS) that continuously learns from routine care data, integrates multiple health data sources, and uses interoperable digital infrastructure (e.g., HL7 and FHIR) to improve antibiotic stewardship and public health reporting. **[4+3+3]**

- Explain how antibiotic resistance can be understood as a systems problem rather than only a biological problem. How would adopting a Learning Health System model help address this issue?
- Explain how different data sources contribute to the feedback loop from data → evidence → improved care.
- Explain how HL7 and FHIR support a Learning Health System by enabling technical and semantic interoperability, and clarify why structured data exchange alone is insufficient without shared clinical meaning.

ANSWER:

a.

Criteria	Marks
Systems perspective	1
Mechanism of resistance emergence	1
Complexity and incomplete knowledge	1
Role of Learning Health System	1

Although resistance originates from bacterial mutations, it is strongly influenced by the complex interactions within healthcare systems. Healthcare systems involve multiple interacting components, including clinicians, prescribing practices, infection control policies, hospital environments, and patient behavior. **Interventions are often based on simplified models that assume “everything else remains equal,” but real-world systems are complex and unpredictable. Therefore, antibiotic resistance becomes a predictable outcome of interventions in complex systems under incomplete knowledge, rather than purely a biological event.**

A Learning Health System (LHS) helps address this problem by **continuously learning from healthcare data. In an LHS, routine care generates data that is analyzed to produce evidence, which is then used to improve care, creating a continuous feedback loop.**

Hospitals can monitor resistance trends, implement antibiotic stewardship strategies, and update clinical policies using real-world data, thereby driving continuous improvement in patient care.

b.

Criteria	Marks
Identification of data sources	1
Data to evidence	1

Evidence to improve care	1
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A Learning Health System integrates multiple healthcare data sources to support continuous learning and improve patient care.

Important sources include **Electronic Health Records (EHRs) containing patient diagnoses and treatments, laboratory information systems providing microbiology and antibiotic susceptibility results, clinical research databases, and public health surveillance datasets** used to monitor disease patterns.

These data sources contribute to a feedback loop:

- **Data collection:** Routine healthcare activities generate large volumes of clinical and laboratory data.
- **Evidence generation:** Data analysis identifies trends such as antibiotic resistance patterns and prescribing behaviors.
- **Improved care:** The generated evidence informs clinical guidelines, decision support systems, and antibiotic stewardship programs.

Through this continuous cycle, healthcare systems learn from real-world practice and improve treatment outcomes.

C.

Criteria	Marks
HL7 explanation	1
FHIR explanation	1
Semantic interoperability	1

- HL7 (Health Level Seven) is a standards development organization that provides messaging standards allowing healthcare systems to exchange clinical information. **It acts as a common language for transmitting healthcare messages between systems such as EHRs, laboratories, and public health databases.**
- FHIR (Fast Healthcare Interoperability Resources) is a modern HL7 standard that enables easier **data exchange using RESTful web APIs and structured healthcare resources. This allows different healthcare applications to share and access information efficiently.**
- However, **technical interoperability alone is insufficient.** While structured data exchange ensures that systems can transmit information, it does not ensure that the systems understand the meaning of the data. **Effective healthcare information exchange requires semantic interoperability, in which systems share a common vocabulary and a consistent interpretation of clinical data. This ensures that the exchanged information can be used accurately for clinical decision-making and public health reporting.**

4. A hospital stores medication entries as free text:
[3+2+2+3]

- “Tab PCM 650”
- “Paracetamol 500 mg”
- “Acetaminophen 1g IV”

You are asked to standardize these using SNOMED CT,

1. Explain how SNOMED CT can normalize these drug entries.
2. What challenges arise in mapping brand names and abbreviations to SNOMED CT concepts?
3. How can SNOMED CT help identify that “Paracetamol” and “Acetaminophen” refer to the same drug?
4. How does using SNOMED CT improve consistency and interoperability between different hospital systems?

Ans SNOMED CT helps convert inconsistent free-text medication entries into standardized clinical concepts using unique identifiers and synonyms. Although challenges exist in mapping abbreviations, brand names and spelling variations, using SNOMED CT greatly improves data consistency, interoperability, and clinical decision support across hospital systems.

a) SNOMED CT assigns a **unique concept identifier** to each clinical concept, including medications. When free-text drug entries are mapped to SNOMED CT:

- “Tab PCM 650”
- “Paracetamol 500 mg”
- “Acetaminophen 1g IV”

can all be mapped to the same **drug concept for paracetamol/acetaminophen**.

The normalization process typically involves:

1. Text processing – extracting the drug name from the free text.
2. Synonym matching – identifying equivalent terms (e.g., PCM = paracetamol).
3. Concept mapping – linking the extracted term to the correct SNOMED CT concept ID.
4. Standard storage – storing the concept ID rather than the free text.

2. Challenges in Mapping Brand Names and Abbreviations

Although SNOMED CT provides standardized terminology, mapping free-text entries to its concepts can present several challenges:

a. Abbreviations

Clinicians often use abbreviations such as “PCM” for paracetamol. These abbreviations may vary across institutions and may not always appear in standardized vocabularies.

b. Brand Names

Many medications are recorded using brand names rather than generic names. A single drug can have multiple brand names, which must be mapped to the correct generic concept.

c. Combination Drugs

Some brand names represent medications with multiple active ingredients, making mapping more complex because each ingredient must be identified.

d. Spelling Variations and Typos

Free-text entries may include spelling errors or variations, which can make automatic mapping difficult.

To address these issues, hospitals often use drug dictionaries, synonym databases, or natural language processing (NLP) tools alongside SNOMED CT.

3. Identifying that “Paracetamol” and “Acetaminophen” Refer to the Same Drug

SNOMED CT organizes medical knowledge using concepts, descriptions, and synonyms. Both “Paracetamol” and “Acetaminophen” are stored as synonymous descriptions linked to the same SNOMED CT concept for the drug substance. Because they share the same concept identifier, any system using SNOMED CT recognizes them as the same medication, even though the names differ.

4. Improving Consistency and Interoperability Between Hospital Systems

a. Consistent Clinical Documentation

All medication entries are standardized using the same concept codes, reducing ambiguity and duplication.

b. Accurate Clinical Decision Support

Standardized drug data enables reliable detection of drug interactions, allergies, and contraindications.

c. Interoperability

When multiple healthcare systems use SNOMED CT, they can exchange medication data seamlessly because the same codes represent the same concepts across systems.

d. Better Data Analysis and Research

Standardized terminology allows researchers to aggregate medication data from different hospitals, improving clinical studies, pharmacovigilance, and healthcare analytics.